Sheng-Chi Hsu

Researcher

Taiwan • ricky610626@gapp.nthu.edu.tw • 0917 803 826 • rickyeeeeee.github.io

Research Interests

I'm interested in *computer graphics*, *3d computer vision*, and *machine learning* research. I have experience in varies *3d reconstruction* topics such as *neural rendering* and *inverse rendering*.

Research Experience

Research Assistant *Hsinchu City, Taiwan*
CPLab, National Yang Ming Chiao Tung University *Sept 2026 – present*
- Advisor: Dr. Yu-Lun Liu at NYCU.
- Working on 3D Generation.

Student Researcher *Hsinchu City, Taiwan*
IGVI Lab, National Tsing Hua University *Sept 2023 – June 2026*
- Advisor: Dr. Hung-Kuo Chu at NTHU.
- 2 papers published.

Education

National Tsing Hua University *Hsinchu City, Taiwan*
MS in Information Systems and Applications *Sept 2024 – present*
- Advisor: Dr. Hung-Kuo Chu at NTHU.
- Research on the novel view synthesis, inverse rendering, and 3d reconstruction algorithms.
- Working on modern graphics API (DirectX 12, Vulkan).

National Tsing Hua University *Hsinchu City, Taiwan*
BS in Electrical Engineering and Computer Science *Sept 2020 – June 2024*
- Undergraduate researcher adised by Prof. Hung-Kuo Chu.

Publications

A2TG: Adaptive Anisotropic Textured Gaussians for Efficient 3D Scene Representation Apr 2026
Sheng-Chi Hsu, Ting-Yu Yen, Shih-Sheng Hung, Hong-Kuo Chu
(*ICLR 2026 conference paper*)

ODA-GS: Occlusion-and Distortion-aware Gaussian Splatting for Indoor Scene Reconstruction Dec 2024
Chai-Rong Lee, Ting-Yu Yen, Kai-Wen Hsiao, Shih-Hsuan Hung, **Sheng-Chi Hsu**, Min-Chun Hu, Chih-Yuan Yao, Hung-Kuo Chu
(*SIGGRAPH ASIA 2024 technical communication*)

Honors and Awards

The 19th Excellent Master Thesis Award (2026): The Chinese Image Processing and Pattern Recognition Society

Projects

Inverse Rendering with Gaussian Splatting *July 2025 – Dec 2025*

A rendering framework for Relightable Gaussian Splatting and PBR mesh hybrid rendering

- Deferred PBR shading for hybrid GS/mesh rendering with post-processing.
- Interactable 3D viewer for real-time manipulation of 3D Gaussians and 3D meshes.

Skills

Programming: Proficient with Python, C++, C, CUDA

Framework: OpenGL, Vulkan, DirectX12, Pytorch, OpenCV

Mathematics: Good understanding of differential equations, calculus, linear algebra, and probability

Languages: Chinese, English